HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Owned the technical direction and architecture for an agentic operating system for global trade, defining the engineering roadmap in collaboration with founders and establishing technical standards for production readiness.
- Designed and implemented durable, long-running AI agents capable of managing complex trade workflows, incorporating state management, resumption capabilities, and dynamic re-planning for changing specifications.
- Developed tool-use capabilities, retrieval systems, and multi-model routing for AI agents, establishing robust evaluation frameworks and release gates for model-driven features within a SaaS platform.
- Built and deployed a secure, multi-tenant SaaS platform using **PostgreSQL** with row-level security and workspace entitlements, defining APIs, data models, and platform extension points.
- Established CI/CD pipelines with **Jenkins** and **Git**, implementing automated testing, monitoring, and rollback processes to ensure platform reliability and performance.
- Embedded security and governance into agent architecture, implementing scoped data access, least-privilege permissions, and prompt-injection defences with a focus on maintaining complete audit trails.
- Managed performance, infrastructure costs, and latency budgets at the workspace level, ensuring predictable scalability for multiple customers, partners, and workflows.

SKILLS

AI/GenAI/LLM – AI Agents, LLM, Tool Use, Retrieval Augmented Generation (RAG), Multi-model Routing, Agent Orchestration, Prompt Injection Defences, Model Evaluation, Fine-tuning, LoRA, SFT, DPO, Unsloth, NVIDIA NeMo, NVIDIA NIM, Model Context Protocol (MCP)

Platform Engineering – SaaS Platform, Multi-tenant Architecture, CI/CD, Automated Testing, Rollback Processes, Performance Management, Infrastructure Cost Management, Latency Budgets

Backend Development – TypeScript, JavaScript, Node.js, Python, Java, Spring Boot, REST APIs, GraphQL APIs, Durable Workflow Engines, Long-running Agent Processes

Databases – PostgreSQL, Row-Level Security, Workspace Entitlements, Partner Permissions, Data Isolation

Cloud & Infrastructure – AWS, Cloud Infrastructure, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, Linux, Shell Scripting

Development Practices – Agile Methodologies, Scrum Ceremonies, Code Reviews, Technical Standards, Production Readiness Criteria, Security-first Engineering, Observability, Incident Management

PROJECTS

Agentic Global Trade Platform Development

- Designed and implemented core agent logic for an AI-powered global trade platform, enabling automated coordination of exporter workflows across partners.
- Developed agent capabilities for state management and resumption, allowing processes to adapt dynamically to changes in dates or rules, ensuring continuous operation over extended periods.
- Integrated retrieval systems and multi-model routing to enhance agent decision-making and tool-use, supporting complex operational workflows for exporters.

Secure SaaS Platform for Trade Operations

- Engineered a multi-tenant SaaS platform architecture with a focus on security and data isolation, implementing granular access controls and permissions.
- Developed REST APIs for seamless integration with external partner systems and defined platform extension points to support future growth and customisation.
- Established robust observability and incident management practices, including detailed audit trails of agent activity and proposed actions for regulatory compliance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)